Refocusing of Moving Ships in Spaceborne SAR Imagery via Space-Variant Phase Compensation

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***Abstract*—Due to the influence of sea waves, winds, and other complex environmental effects, maritime ships often suffer from severe defocusing in high-resolution synthetic aperture radar (SAR) images. The motions of the ship typically exhibit distinct space-variant features, which pose great challenges in ship imaging. To correct this spatial variance, this article proposes a space-variant phase compensation (SVPC) method based on the analysis of composite motion coupling between translational and rotational motions. Specifically, after the image segmentation, we first present a fast iterative algorithm based on monotonic iterative minimum entropy (MIME) to compensate for the azimuth-invariant phase terms. It applies an optimized transformation technique to achieve rapid indirect descent of the objective function. Then, we propose an azimuth-variant phase compensation algorithm based on the nonlinear chirp scaling (NCS) principle. It suppresses azimuthal variance through the perturbation function and modified Doppler compensation and, finally, utilizes a consistent compression filter to refocus the azimuth signal. Ultimately, experiments on both simulated and real SAR data demonstrate the effectiveness and robustness of the proposed method, outperforming other comparison techniques.**

***Index Terms*—Refocusing, ship imaging, space-variant phase compensation (SVPC), synthetic aperture radar (SAR).**

I. Introduction

SPACEBORNE synthetic aperture radar (SAR) is an effective remote sensing detector with all-day and all-weather imaging capability and has been widely applied in fields such as ground reconnaissance, environmental remote sensing, and disaster prevention [1], [2], [3]. One of its most significant applications is maritime surveillance, where spaceborne SAR sensors can make use of their advantages of wide-swath coverage and high resolution to obtain high-quality imagery for ship recognition and interpretation. However, under the influence of sea waves and winds, maritime ships are often subject to rotational motions (i.e., roll, pitch, and yaw) [4], [5], [6]. At the same time, ships also undergo translational motion along their routes, which couples with the rotational motions, as illustrated in Fig. 1. These nonstationary motions pose significant challenges to traditional SAR imaging algorithms, often resulting in defocusing and blurring, reducing the image quality, and limiting the subsequent processing. Therefore, with the development of high-resolution SAR technology, refocusing of moving ships has become increasingly important and is attracting growing attention.

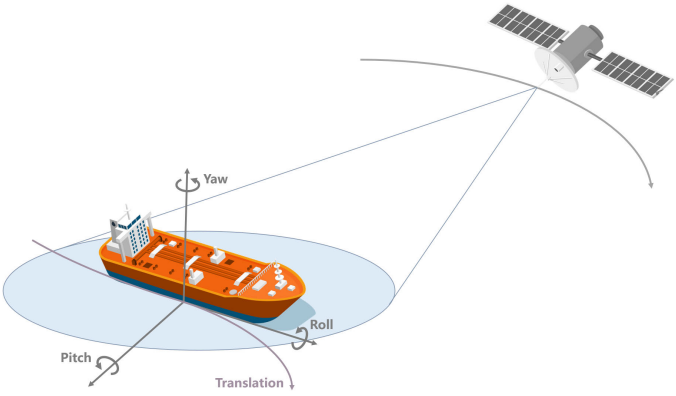


Fig. 1. SAR imaging of a moving ship with translational and rotational motions.

In spaceborne SAR systems, due to the relatively remote distance between the satellite and the ship, the variation in slant range history caused by complex motion would primarily affect the phase of the echo, rather than causing significant range cell migration. Therefore, most defocused targets in SAR imagery exhibit severe azimuthal blurring, where incomplete phase compensation leads to azimuthal mismatch. Currently, established refocusing methods for ships are almost entirely based on phase estimation or azimuthal processing and can generally be divided into two categories: SAR-based and inverse SAR (ISAR)-based methods.

For SAR-based methods, they can be further categorized into nonparametric and parametric methods. The most classical nonparametric method is phase gradient autofocus (PGAs) along with its modified versions [7], [8], [9], [10]. PGA-based methods, on the basis of a statistical model rather than a phase model, can unbiasedly estimate the phase gradient in azimuth and iteratively compensate for the errors, achieving good performance on translational motion compensation. The metric-based methods represent another mature nonparametric technology. These methods optimize image quality metrics, such as image entropy, contrast, and sharpness [11], [12], [13], [14], and directly estimate azimuth phase errors, thus offering considerable robustness. As for parametric methods, map drift

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(MD) is a classical technology [15], [16] which is based on a phase model and estimates the Doppler modulated rate (DMR) by cross correlation operation between subapertures. Recently, some studies have modeled the defocused signal as a multicomponent-linear frequency modulated (MC-LFM) signal and applied the MC-LFM-based methods to estimate the signal parameters, thereby reconstructing refocused azimuthal lines [17], [18], [19], [20]. Beyond these, some methods are built upon classical SAR imaging algorithms, embedding motion parameters and making estimation by image quality optimization [21], [22], [23], [24]. Most SAR-based methods have a good performance on refocusing translationally moving ships with constant motion characteristics, but they often struggle to handle the severe spatial variance caused by nonstationary motion.

ISAR-based methods are also referred to as SAR/ISAR hybrid imaging. As the name suggests, the coarse SAR imaging results are mapped into the ISAR data domain for further processing. After translational motion compensation [25], [26], range Doppler (RD) or range instantaneous Doppler (RID) imaging is applied to refocus the rotating target. In the case of RD imaging, some studies choose to estimate the optimal subaperture and imaging time, which can lead to low resolution under nonuniform rotation [27], [28], [29], [30], [31]. Meanwhile, other works apply the variant phase coefficients' estimation to overcome the spatial variance coupling in the aperture, thereby achieving full aperture imaging [32], [33], [34], [35], [36]. Regarding RID imaging, since the translational-motion-compensated ISAR data exhibit a Doppler bandwidth caused by high-order rotation, it utilizes the time–frequency distribution to directly obtain the instantaneous Doppler and chooses the optimal time samples to reconstruct each azimuthal line [37], [38]. Although RID imaging avoids complex motion estimation, it faces a tradeoff between cross-term accumulation and resolution. There are also some MC-LFM-based methods applied in ISAR imaging to reconstruct the time–frequency distribution without cross-term accumulation [39], [40], [41].

To address the issues, this article approaches the problem from the perspective of a space-variant polynomial phase model of moving ships and proposes a novel refocusing method based on space-variant phase compensation (SVPC). The main innovations are given as follows.

1) We construct a composite slant range model that combines translational slant range with rotational variation. By further deriving the phase model, we decompose it into azimuth-variant and azimuth-invariant terms and design compensation strategies for each term.
2) To eliminate the azimuth-invariant phase terms, we present a fast iterative algorithm based on monotonic iterative minimum entropy (MIME). The algorithm employs an optimized transformation technique to achieve precise indirect descent of the objective function.
3) For the azimuth-variant phase terms, we propose a subimage-processing algorithm based on the nonlinear chirp scaling (NCS) principle. The algorithm robustly suppresses the azimuth-variant features through a high-order perturbation function and modified Doppler compensation. Finally, we apply consistent azimuth compression to realize refocusing.

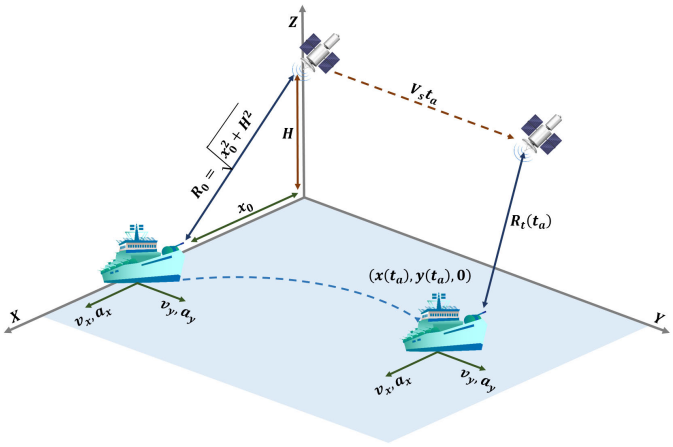


Fig. 2. Imaging geometry of the ship with translational motion.

The rest of this article is organized as follows. Section II introduces the signal model and phase analysis for moving ships. Section III proposes the refocusing method based on SVPC according to the features of the phase model. Section IV demonstrates the effectiveness of the proposed method through numerical simulation and real SAR data. Ultimately, Section V concludes this article.

II. Imaging Geometry and Phase Analysis

A moving ship on the sea surface generally undergoes a complex motion that combines translation and rotation, which poses challenges in the consistent analysis of the slant range model. Therefore, we propose to analyze the translation and rotation separately and then combine the two parts to form a composite slant range model.

A. Translational Slant Range Model

Consider a spaceborne SAR imaging scenario involving a ship in pure translational motion. The SAR sensor is working in the side-looking mode, as illustrated in Fig. 2, and the direction of the beam is approximately perpendicular to the flight trace, in which the incidence angle of the radar is θ_n. A coordinate system O-XYZ is defined relative to the sea surface, where the Z-axis is perpendicular to the sea surface, the Y-axis points along the satellite's flight (azimuth) direction, and the X-axis represents the ground-range direction. The equivalent ground velocity of low-orbital spaceborne SAR is V_s and its orbital height is H. At an instantaneous azimuthal time t_a, the position coordinates of a target are given by $(x(t_a), y(t_a), 0)$. The ship's motion is modeled with second-order dynamics in both ground-range and azimuth directions, which is consistent for all the points on the ship. In addition, we assume that the minimum slant range is $R_0 = (x_0^2 + H^2)^{1/2}$, where x_0 represents the minimum ground range at the aperture center time t_0.

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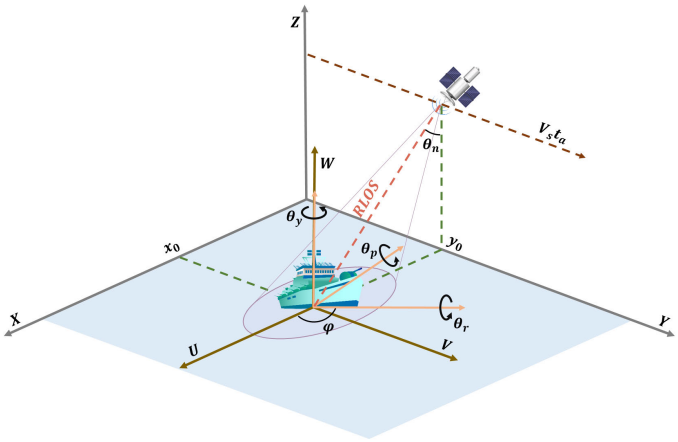


Fig. 3. Imaging geometry of the ship with rotational motion.

Based on the imaging geometry, the instantaneous slant range for any scattering point on the translational target can be calculated as

$$R_t(t_a) = \sqrt{[x(t_a)]^2 + \left[V_s t_a - y(t_a)\right]^2 + H^2}. \tag{1}$$

Then, at the synthetic aperture center t_0, the third-order Taylor series expansion of the above equation is applied to obtain its cubic polynomial form as follows:

$$R_t(t_a) = R_0 + K_1(t_a - t_0) + K_2(t_a - t_0)^2 + K_3(t_a - t_0)^3 \tag{2}$$

$$\begin{cases} K_1 = \dfrac{v_x x_0}{R_0} = v_x \sin\theta_n \\ K_2 = \dfrac{1}{2R_0}\left[\left(V_s - v_y\right)^2 + v_x^2\cos^2\theta_n + x_0 a_x\right] \\ K_3 = \dfrac{1}{2R_0^2}\left\{\begin{array}{l} R_0\left[a_x v_x - a_y\left(V_s - v_y\right)\right] - v_x\sin\theta_n \\ \cdot\left[\left(V_s - v_y\right)^2 + v_x^2\cos^2\theta_n + x_0 a_x\right] \end{array}\right\}. \end{cases} \tag{3}$$

According to (2) and (3), it can be observed that the second- and third-order terms in the translational slant range model are influenced by range-dependent parameters, such as the minimum slant range and ground range. However, due to the limited variation in the minimum slant range and generally small acceleration, the spatial variance is relatively small in the translational motion of the ship, and many refocusing methods can achieve good results.

In addition, the first-order term in (2) represents the Doppler center shift of the ship in SAR imagery, which is caused by the radial velocity and is independent of spatial position. Therefore, some repositioning approaches rely on precise estimation of the radial velocity to correct this shift.

B. Rotational Slant Range Variation Model

Consider the imaging geometry for a rotating ship, as shown in Fig. 3. The parameters of the radar system are identical to those in Section II-A. Except for the coordinate system *O-XYZ* of the sea surface, we also establish the ship-fixed coordinate system *O′-UVW*, with its origin located at the ship center and its axes parallel to the corresponding axes of the sea surface coordinate system. At the time of the synthetic aperture center t_0, the radar line of sight (RLOS) is directed to the centroid of the ship target on the sea surface. φ represents the angle of ship heading and the ship undergoes rotational motion with three degrees of freedom in the ship-fixed coordinate system, that is, roll, pitch, and yaw, where θ_r, θ_p, and θ_y denote the roll, pitch, and yaw angles, respectively.

For a given scattering point P, we assume that its initial coordinates in the ship-fixed coordinate system are (u_0, v_0, w_0). By analyzing the rotational motion during the synthetic aperture period, we could obtain the instantaneous position $(u_r(t_a), v_r(t_a), w_r(t_a))$ as follows:

$$\begin{bmatrix} u_r(t_a) \\ v_r(t_a) \\ w_r(t_a) \end{bmatrix} = \mathbf{R}_W(-\varphi)\,\mathbf{R}_U(\theta_r)\,\mathbf{R}_V(\theta_p)\,\mathbf{R}_W(\theta_y)\,\mathbf{R}_W(\varphi) \begin{bmatrix} u_0 \\ v_0 \\ w_0 \end{bmatrix} \tag{4}$$

where $\mathbf{R}_U, \mathbf{R}_V$, and $\mathbf{R}_W$, respectively, represent the rotational matrices about the *U*-, *V*-, and *W*-axes in the ship-fixed coordinate system. Under the reasonable small-angle approximations to trigonometric functions and ignorance of the initial height of P, we could further derive the specific forms

$$\begin{cases} u_r = \left[(u_0\cos\varphi + v_0\sin\varphi) + (u_0\sin\varphi - v_0\cos\varphi)\theta_y\right]\cos\varphi \\ \quad - \left[(u_0\cos\varphi + v_0\sin\varphi)\theta_y \right. \\ \quad\quad \left. - (u_0\sin\varphi - v_0\cos\varphi)\right]\sin\varphi \\ v_r = \left[(u_0\cos\varphi + v_0\sin\varphi) + (u_0\sin\varphi - v_0\cos\varphi)\theta_y\right]\sin\varphi \\ \quad + \left[(u_0\cos\varphi + v_0\sin\varphi)\theta_y \right. \\ \quad\quad \left. - (u_0\sin\varphi - v_0\cos\varphi)\right]\cos\varphi \\ w_r = (u_0\cos\varphi + v_0\sin\varphi)\theta_p - (u_0\sin\varphi - v_0\cos\varphi)\theta_r. \end{cases} \tag{5}$$

According to (5), the rotational coordinates vary with the rotational angles. We usually model the angles as oscillating variations using trigonometric functions

$$\begin{cases} \theta_r(t_a) = A_r\sin(2\pi t_a/T_r + \phi_r) \\ \theta_p(t_a) = A_p\sin\left(2\pi t_a/T_p + \phi_p\right) \\ \theta_y(t_a) = A_y\sin\left(2\pi t_a/T_y + \phi_y\right) \end{cases} \tag{6}$$

where r, p, and y represent roll, pitch, and yaw, respectively, i.e., three degrees of freedom in rotation. Since the rotational periods of the ship (typically around 10 s) are much longer than the synthetic aperture period of low-orbital spaceborne SAR, we can perform a polynomial expansion on (6) to fit the variation of the rotational angles in the analysis

$$\theta_k(t_a) = h_0^k + h_1^k t_a + h_2^k t_a^2 + h_3^k t_a^3, \quad k = r, p, y \tag{7}$$

$$\begin{cases} h_0^k = A_k\sin\phi_k \\ h_1^k = \dfrac{2\pi}{T_k}A_k\cos\phi_k \\ h_2^k = -\left(\dfrac{2\pi}{T_k}\right)^2\dfrac{A_k}{2}\sin\phi_k \\ h_3^k = -\left(\dfrac{2\pi}{T_k}\right)^3\dfrac{A_k}{6}\cos\phi_k. \end{cases} \tag{8}$$

To analyze the additional slant range variation induced by the ship's rotational motion under the far-field assumption, we

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project the position variation before and after rotation onto the RLOS direction, defined as the rotational slant range variation as

$$\Delta R(t_a) = \mathbf{\Delta P} \cdot \mathbf{L}_{RLOS} = \begin{bmatrix} u_r - u_0 \\ v_r - v_0 \\ w_r - w_0 \end{bmatrix} \cdot \begin{bmatrix} \sin\theta_n \\ 0 \\ -\cos\theta_n \end{bmatrix}. \tag{9}$$

Combining (5) and (7)–(9), we could further derive the specific polynomial form of rotational slant range variation

$$\begin{aligned} R_r(t_a) &= \Delta R(t_a - t_0) \\ &= \left(K_0^u + K_0^v\right) + \left(K_1^u + K_1^v\right)(t_a - t_0) \\ &\quad + \left(K_2^u + K_2^v\right)(t_a - t_0)^2 + \left(K_3^u + K_3^v\right)(t_a - t_0)^3 \end{aligned} \tag{10}$$

$$\begin{cases} K_i^u = u_0 \cos\theta_n \left(h_i^r \sin\varphi - h_i^p \cos\varphi\right) \\ K_i^v = -v_0 \left[h_i^y \sin\theta_n + \cos\theta_n \left(h_i^r \cos\varphi + h_i^p \sin\varphi\right)\right] \end{cases} \tag{11}$$

where $i = 0, 1, 2, 3$ in (11). Based on this derivation, it can be observed that the rotational motion affects the slant range with spatial variance in two dimensions, i.e., K_i^u and K_i^v in each order term, which are, respectively, related to the ground range and azimuth position.

C. Signal Model and Space-Variant Phase

For a spaceborne radar system on a low-orbital satellite platform, the beam irradiates the ship target, and the backscattered signal is received by the system for processing. The range-compressed signal of each scattering point on the target could be expressed as

$$\begin{aligned} s_m(t, t_a) &= \sigma_0 w_a \left(\frac{t_a - t_0}{T_a}\right) \sin c\left\{B\left(t - \frac{2R(t_a)}{c}\right)\right\} \\ &\quad \cdot \exp\left\{-j\frac{4\pi}{\lambda} R(t_a)\right\} \end{aligned} \tag{12}$$

where t represents the fast time, σ_0 represents the backscattering coefficient, c is the light speed, and λ denotes the wavelength of the electromagnetic wave.

Since ship motion has only weak effects on range processing, that is, the extra range cell migration caused by nonstationary motion is relatively small compared to that induced by the SAR platform, we can perform range cell migration correction (RCMC) and obtain the range-processed signal as follows:

$$\begin{aligned} s_p(t, t_a) &= \sigma_0 w_a \left(\frac{t_a - t_0}{T_a}\right) \sin c\left\{B\left(t - \frac{2R_0}{c}\right)\right\} \\ &\quad \cdot \exp\left\{-j\frac{4\pi}{\lambda} R(t_a)\right\} \end{aligned} \tag{13}$$

where T_a represents the synthetic aperture period and the processing of RCMC can be referred to in the RD algorithm (RDA). After the range processing, the core challenge in ship imaging actually lies in phase compression in the azimuth direction. Therefore, we can consider (13) as the azimuth signal in the following form:

$$s_p(t_a) = \sigma_0 w_a \left(\frac{t_a - t_0}{T_a}\right) \exp\left\{-j\frac{4\pi}{\lambda} R(t_a)\right\}. \tag{14}$$

$R(t_a)$ in the phase term of the azimuthal signal represents the composite slant range, composed of both translational and rotational motions. Based on the analysis above, we can derive the phase terms in (14) as

$$\begin{aligned} &\phi(t_a) \\ &= -\frac{4\pi}{\lambda}\left[R_t(t_a) + R_r(t_a)\right] \\ &= -\frac{4\pi}{\lambda}\begin{pmatrix} R_0 + K_1(t_a - t_0) + K_2(t_a - t_0)^2 + K_3(t_a - t_0)^3 \\ + \left(K_0^u + K_0^v\right) + \left(K_1^u + K_1^v\right)(t_a - t_0) \\ + \left(K_2^u + K_2^v\right)(t_a - t_0)^2 + \left(K_3^u + K_3^v\right)(t_a - t_0)^3 \end{pmatrix}. \end{aligned} \tag{15}$$

For the azimuth-variant terms in (15), in fact, the coefficients K_i^v related to the azimuthal position could be relevant with the synthetic aperture center of each scattering point and modeled as first-order spatial variance

$$K_i^v = k_{0,i} + t_0 k_{1,i}. \tag{16}$$

By combining (15) and (16) and simplifying like terms, we can formulate the polynomial phase model. Specifically, the complex amplitude introduces a fixed phase shift to the signal and may influence the coherence and scattering features of the image. The linear phase terms correspond to the shift of the Doppler center caused by the radial motion of the target (including both translational and rotational components), and they cause an azimuth shift of the target in the SAR image. They are irrelevant to defocusing and energy spreading. Ignoring the complex amplitude and the linear terms, we can obtain the space-variant phase as follows:

$$\begin{aligned} \phi_p(t_a) &= \phi_a(t_a) + \phi_r(t_a) \\ &= \pi t_0 A_2 (t_a - t_0)^2 + \pi t_0 A_3 (t_a - t_0)^3 \\ &\quad + \pi B_2 (t_a - t_0)^2 + \pi B_3 (t_a - t_0)^3 - \pi K_a (t_a - t_0)^2 \end{aligned} \tag{17}$$

where

$$\begin{cases} A_2 = (-4/\lambda) \cdot k_{1,2} \\ A_3 = (-4/\lambda) \cdot k_{1,3} \\ B_2 = (-4/\lambda) \cdot \left(k_{0,2} + K_2 + K_2^u\right) + K_a \\ B_3 = (-4/\lambda) \cdot \left(k_{0,3} + K_3 + K_3^u\right) \end{cases} \tag{18}$$

and K_a represents the azimuthal chirp rate for a stationary target, e.g., its value can be approximately calculated as $K_a = 2V_s^2/\lambda R_0$ in RDA.

It can be seen that the phase terms in (17) are divided into azimuth-variant and azimuth-invariant terms, which realizes the translation from nonstationary motion to space-variant phase. Conventional SAR imaging algorithm, such as RDA, utilizes an azimuthal filter with parameters for a stationary target and cannot make complete compression for these phase terms. Consequently, space-variant phase error will degrade imagery quality and cause space-variant defocusing in different sections of the ship target.

III. Refocusing Method

In this section, we propose a ship target refocusing method based on SVPC. First, to mitigate the severe space-variant degradation in defocused ship imagery, we perform image segmentation and use entropy-based criteria to extract subimages

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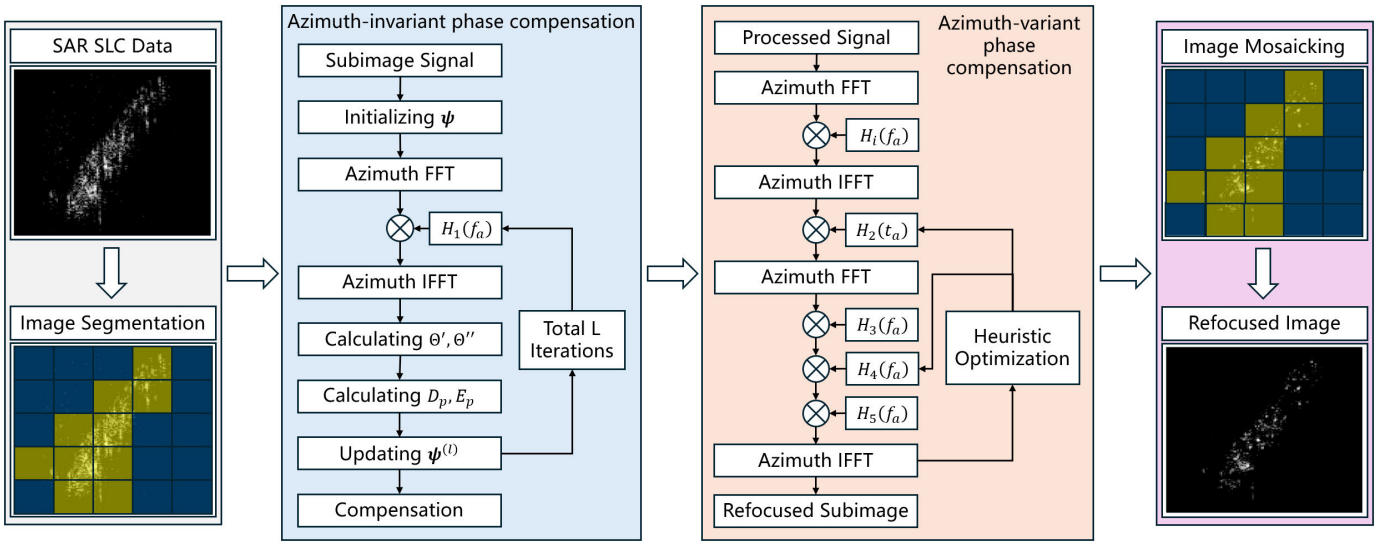


Fig. 4. Flowchart of the refocusing method based on SVPC.

containing the target. Subsequently, based on the azimuthal characteristics analyzed above, we decompose the phase compensation task into two components. We first introduce an azimuth-invariant phase compensation algorithm based on MIME, which enables fast and accurate iterative parameter estimation for the constructed third-order compensation filter. Following this, we implement an azimuth-variant phase compensation algorithm based on the NCS principle. This algorithm incorporates time-domain NCS and Doppler analysis to robustly eliminate the azimuth-variant features and, finally, obtain the refocused target by image stitching. The overall processing flow of the proposed method is illustrated in Fig. 4.

A. Imagery Segmentation and Discrimination

Due to the strong space-variant defocusing across different target components, most frequency-domain algorithms are affected, which, in severe cases, can even limit refocusing performance. Therefore, we perform image segmentation to further mitigate the influence of strong spatial variance.

For an image chip containing a detected ship target in SAR imagery, we generally divide it into segments along the azimuth direction, taking into account both the ship's size and the distribution of strong scattering points. Overlapping areas are set between adjacent segments, accounting for a certain proportion to guarantee the continuity of the corrected image. Sometimes, the defocusing feature is dominated by range variance under visual inspection, and we can appropriately reduce the number of azimuth segments because the azimuth-variant phase compensation is sufficient to compensate for the subsidiary azimuth variance. In the range direction, segmentation is carried out as finely as possible without compromising the continuity of adjacent ship components. Ultimately, subimages are naturally formed by the grid obtained from the azimuth and range segmentation, as shown in Fig. 5(a).

After the initial partitioning, the SVPC method is applied to each subimage as described later. We adjust the scale

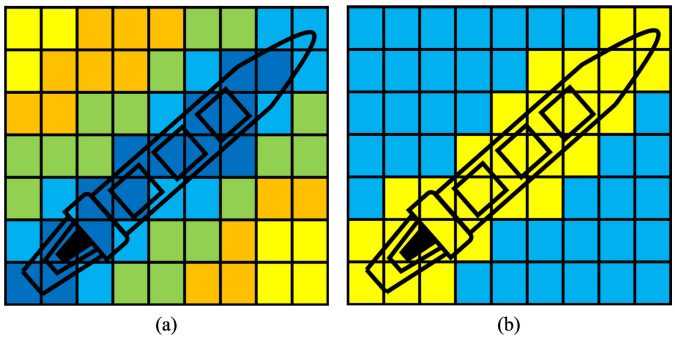


Fig. 5. Imagery segmentation and discrimination via image entropy for a ship target with sea clutter. (a) Distribution of image entropy for the ship, where the colors from blue to yellow represent the value of entropy from low to high. (b) Signal discrimination for each subimage, where the yellow colors denote classification as the target, and the blues denote classification as the background.

of subimages from sparse to dense in order to improve the compensation effect. Ultimately, we can determine the final segmentation strategy, which yields the best-refocused results.

Regarding the discrimination of target energy, our approach is based on an observed phenomenon in real data: some subimages contain only sea clutter energy due to the specific orientation of the ship. Refocusing these subimages would not only limit processing effectiveness but also degrade the refocused quality of the ship target. Thus, to distinguish between target and sea clutter, we utilize subimage entropy as the discriminating criterion

$$IE(s) = -\frac{1}{E}\sum_{m,n}|s(m,n)|^2 \ln|s(m,n)|^2 + \ln E \tag{19}$$

$$E = \sum_{m,n}|s(m,n)|^2. \tag{20}$$

Different from the aggregation of the ship signal, sea clutter exhibits a more dispersed distribution and corresponds to a higher value of image entropy. Thus, we can set a normalized threshold τ and filter out the subimages containing the ship

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target, as shown in Fig. 5(b)

$$Mask(i) = \begin{cases} 0, & IE(i) > \tau \cdot \max\{IE\} \\ 1, & IE(i) \le \tau \cdot \max\{IE\}. \end{cases} \tag{21}$$

The range of τ depends on the size of the chip and the posture of the ship, and is generally set to $\tau \ge 0.7$ in most cases.

B. Azimuth-Invariant Phase Compensation

After image segmentation of the SAR image, we first consider the azimuth-invariant phase in the range-processed signal

$$s_r(t_a) = \sigma_0 w_a\left(\frac{t_a - t_0}{T_a}\right)\exp\left\{j\left[\pi B_2 (t_a - t_0)^2 + \pi B_3 (t_a - t_0)^3\right]\right\}. \tag{22}$$

Owing to the invariant features of the coefficients in the phase terms in (22), it can be recognized that the Doppler histories of individual point scatterers are similar, thus allowing consistent processing. Applying the principle of stationary phase (POSP) and the method of series reversion (MSR), we can obtain the Doppler spectrum of the above signal as follows:

$$S_r(f_a) = \sigma_0 W_a(f_a)\exp\left\{j\left(2\pi f_a t_0 + \frac{\pi}{B_2} f_a^2 - \frac{\pi B_3}{B_2^3} f_a^3\right)\right\}. \tag{23}$$

The spectrum above exhibits a polynomial form for each point scatterer and has been translated to zero Doppler. Thus, a third-order compensation filter can be constructed in the Doppler domain

$$H_1(f_a) = \exp\left\{j\pi\left(\psi_2 f_a^2 + \psi_3 f_a^3\right)\right\}. \tag{24}$$

Therefore, letting $I_r(m,n)$ be the refocused signal, with m and n representing the sample index of azimuth and range, respectively, the processing of azimuth-invariant phase compensation can be modeled as the following optimization problem:

$$\hat{\boldsymbol{\psi}} = \arg\min\{IE(I_r(m,n);\boldsymbol{\psi})\} \tag{25}$$

where $\boldsymbol{\psi} = (\psi_2, \psi_3)$ represents the parameters vector of azimuth-invariant compensation to be estimated.

To achieve a quick solution of parameters, the Doppler spectrum in (23) and the compensation filter in (24) are discretized, yielding the discrete refocused signal as

$$\begin{aligned} I_r(m,n) &= \mathrm{IFT}\{S_r(k,n)\cdot H_1(k)\} \\ &= \sum_k S_r(k,n)\cdot\exp\{j\xi(k)\}\cdot\exp\left\{j2\pi\frac{km}{M}\right\} \end{aligned} \tag{26}$$

where IFT$\{\cdot\}$ denotes the inverse Fourier transform (IFT), k denotes the indexes of azimuth samples in the frequency domain, $S_r(k,n)$ denotes the discrete Doppler spectrum, and $\xi(k)$ denotes the phase of the compensation filter

$$\xi(k) = \psi_2 f_a^2(k) + \psi_3 f_a^3(k). \tag{27}$$

In fact, (25) does not have a closed-form solution. Thus, we must resort to iterative algorithms. Unlike common gradient-based techniques or successive approaching strategies, the MIME algorithm employs optimized transformation techniques that realize indirect descent of the objective function via gradient descent of a surrogate function [11]. Therefore, the surrogate function can serve as an upper bound on the objective function, and according to former works, one suitable form of surrogate function could be expressed as follows, referred to (19):

$$\boldsymbol{\psi} = \arg\min\{IE(I_r;\boldsymbol{\psi})\} = \lim_{l\to\infty}\arg\min\left\{\Theta\left(\boldsymbol{\psi};\boldsymbol{\psi}^{(l)}\right)\right\} \tag{28}$$

$$\Theta\left(\boldsymbol{\psi};\boldsymbol{\psi}^{(l)}\right) = -\frac{1}{E}\sum_{m,n}|I_r|^2\ln\left|I_r^{(l)}\right|^2 + \ln E \tag{29}$$

where $\boldsymbol{\psi}^{(l)}$ represents the parameter estimation after the lth iteration and $I_r^{(l)}$ denotes the refocused signal compensated by $\boldsymbol{\psi}^{(l)}$.

Due to the challenge of multiparameter optimization, we adopt the coordinate descent principle to (29), wherein the surrogate function reduces to a scalar function

$$\psi_p^{(l+1)} = \arg\min\Theta\left(\psi_p;\boldsymbol{\psi}^{(l)}\right). \tag{30}$$

In addition, under the condition of coordinate descent, the scalar surrogate function possesses the property that its first- and third-order derivatives have opposite signs at the position $\psi_p = \psi_p^{(l)}$. Hence, the analytic form of the scalar surrogate function can be expressed as follows:

$$\begin{aligned} \Theta\left(\psi_p;\boldsymbol{\psi}^{(l)}\right) &= D_p\cos\left(\psi_p - \psi_p^{(l)}\right) \\ &\quad + E_p\sin\left(\psi_p - \psi_p^{(l)}\right) + F_p \end{aligned} \tag{31}$$

$$\begin{cases} D_p = -\dfrac{d^2\Theta}{d\psi_p^2}\Big|_{\psi_p=\psi_p^{(l)}} \\ E_p = \dfrac{d\Theta}{d\psi_p}\Big|_{\psi_p=\psi_p^{(l)}} \\ F_p = \Theta\left(\psi_p;\psi_p^{(l)}\right) + \dfrac{d^2\Theta}{d\psi_p^2}\Big|_{\psi_p=\psi_p^{(l)}}. \end{cases} \tag{32}$$

Combining (26), (29), and (32), we obtain the analytic form of each derivative as follows:

$$\begin{cases} \dfrac{\partial\Theta}{\partial\psi_p} = -\dfrac{1}{E}\displaystyle\sum_{m,n}2\mathrm{Re}\left[I_r^*\cdot\frac{\partial I_r}{\partial\psi_p}\right]\ln\left|I_r^{(l)}\right|^2 \\ \dfrac{\partial I_r}{\partial\psi_p} = \dfrac{j}{M}\displaystyle\sum_k S_r(k,n)\left[f_a(k)\right]^p\exp\{j\xi(k)\}\exp\left\{j2\pi\frac{km}{M}\right\} \\ \dfrac{\partial^2\Theta}{\partial\psi_p^2} = -\dfrac{1}{E}\displaystyle\sum_{m,n}2\left\{\left|\frac{\partial I_r}{\partial\psi_p}\right|^2 + \mathrm{Re}\left[I_r^*\cdot\frac{\partial^2 I_r}{\partial\psi_p^2}\right]\right\}\ln\left|I_r^{(l)}\right|^2 \\ \dfrac{\partial^2 I_r}{\partial\psi_p^2} = -\dfrac{1}{M}\displaystyle\sum_k S_r(k,n)\left[f_a(k)\right]^{2p}\exp\{j\xi(k)\} \\ \qquad\cdot\exp\left\{j2\pi\frac{km}{M}\right\}. \end{cases} \tag{33}$$

While (33) provides a closed-form solution, our primary interest lies in the phase within the interval $[-\pi,\pi]$. By setting the derivative to zero in (31), we consequently obtain the following iterative expression:

$$\psi_p^{(l+1)} = \psi_p^{(l)} + \arctan\left(\frac{E_p}{D_p}\right). \tag{34}$$

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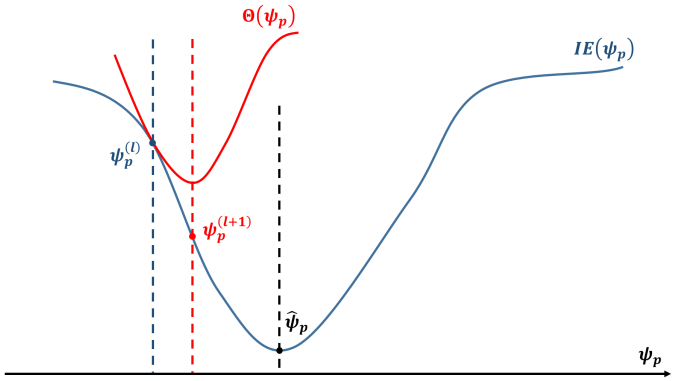


Fig. 6. Schematic of iteration in MIME, where the blue line represents the objective function (i.e., image entropy function), and the red line represents the scalar surrogate function.

By solving for the parameters in the above equations, we can directly compute the extremum of the surrogate function at each iteration, thereby achieving an indirect descent of the objective function across multiple iterations.

In summary, MIME transforms the nonconvex minimum entropy optimization problem into a sequence of surrogate-function optimization problems, as illustrated in Fig. 6. By analytically solving for the extremum of the surrogate function at each iteration, MIME achieves adaptive step-size adjustment, leading to faster convergence and improved robustness compared with conventional gradient-based optimization methods. In addition, the FFT-based gradient computation and the low-order polynomial phase model further reduce the computational complexity.

We assume that each range block has M azimuth samples and N range samples. There are four parameters with complex forms, and fortunately, we can utilize matrix computation and fast Fourier transform (FFT) to accelerate the computation. Since complex multiplication is typically the most time-consuming operation, we use its count to represent the computational cost of MIME. The computational load of the first-order derivative for I_r is $O((p+1)MN + MN\log M/2)$, and that of the second-order derivative is $O((2p+1)MN + MN\log M/2)$. Similarly, the computational load of the first-order derivative for Θ is $O(3MN)$, and the second-order derivative is $O(4MN)$. Overall, the computational cost per iteration is the sum of the aforementioned quantities, i.e., $O(3(p+3)MN + MN\log M)$. Since we only have two parameters to estimate, i.e., $p = 2$ or $p = 3$, we combine the two parts of quantities, and the estimation is $O(33MN + MN\log M)$. Consequently, after L iterations for B blocks, the total computational cost is $O(33BLMN + BLMN\log M)$. Given that the numbers of iterations L and blocks B are typically much smaller than the size of the SAR image, the overall computational cost remains low (specifically less than fourth power magnitude), ensuring the efficiency of the fast algorithm.

C. Azimuth-Variant Phase Compensation

After the azimuth-invariant compensation, the residual phase exhibits azimuth-variant features, posing significant challenges for classical algorithms. The NCS algorithm was originally proposed for squinted SAR and is also capable of addressing the challenges caused by azimuthal variance [23], [33], [42]. The method employs a perturbation function to overcome the variance of frequency modulation for point scatterers at different spatial positions. In this section, we propose a compensation method based on the NCS principle and apply the modified method to further compress the variance in the range-processed signal, which can be utilized both in SAR imaging and postprocessing.

According to the NCS principle, variant chirp rates are balanced to basic frequency modulation via a perturbation function. For the range-processed signal in (17), the original chirp rate K_a can be used as the basic chirp rate. In the postprocessing of the coarse SAR image, the range-processed signal with chirp rate can be obtained via inverse azimuth compression as follows:

$$H_i(f_a) = \exp\left\{j\frac{\pi}{K_a}f_a^2\right\} \tag{35}$$

where f_a denotes the azimuth frequency. Overall, we can finally express the range-processed signal with azimuthal variance for each point scatterer in each subimage as

$$s_a(t_a) = \sigma_0 w_a\left(\frac{t_a - t_0}{T_a}\right)\exp\left\{j\pi\left[-K_a(t_a-t_0)^2 + t_0A_2(t_a-t_0)^2 + t_0A_3(t_a-t_0)^3\right]\right\}. \tag{36}$$

In order to correct the second- and third-order azimuth-variant phases, we can utilize a fourth-order perturbation function to multiply the above signal in the time domain. The perturbation function can be expressed as follows:

$$H_2(t_a) = \exp\left\{j\pi\left(\alpha t_a^3 + \beta t_a^4\right)\right\}. \tag{37}$$

After multiplication with the perturbation function and neglecting the complex amplitude of the perturbed signal $s_{a1}(t_a)$, we can reorganize the signal in the time domain as

$$\begin{aligned} s_{a1}(t_a) = \sigma_0 w_a\left(\frac{t_a - t_0}{T_a}\right)&\exp\left\{j\pi\left(3\alpha t_0^2 + 4\beta t_0^3\right)(t_a-t_0)\right\} \\ &\cdot\exp\left\{j\pi\left(t_0A_2 + 3\alpha t_0 + 6\beta t_0^2\right)(t_a-t_0)^2\right\} \\ &\cdot\exp\left\{j\pi\left(t_0A_3 + 4\beta t_0\right)(t_a-t_0)^3\right\} \\ &\cdot\exp\left\{j\left[-\pi K_a(t_a-t_0)^2 + \pi\alpha(t_a-t_0)^3 + \pi\beta(t_a-t_0)^4\right]\right\}. \end{aligned} \tag{38}$$

According to the above equation, the phase terms can be divided into three categories: the first phase term represents the extra Doppler center, the second and third terms denote the perturbed phases, and the last term is the invariant phase term. Thus, to eliminate the perturbed phase terms, we can derive the conditions for the double-scaling factors as

$$\begin{cases} \hat{\alpha} = -\dfrac{A_2}{3} - 2\hat{\beta}t_0^{ref} \\ \hat{\beta} = -\dfrac{A_3}{4} \end{cases} \tag{39}$$

where t_0^{ref} represents the referred aperture center. Except for the perturbed phase, the remaining high-order terms are invariant and can be corrected using known parameters. Furthermore, the above conditions are related to the synthetic

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aperture center, and processing of each subimage with respect to its aperture center helps reduce the processing error in different aperture centers of long-distance scatterers.

Assuming that the scaling factors have been estimated accurately, we now further apply the POSP and MSR to derive the spectrum of the NCS-processed signal

$$\begin{aligned} S_{a2}(f_a) = {} & \sigma_0 W_a(f_a - f_{d0}) \exp\{-j2\pi(f_a - f_{d0})t_0\} \\ & \cdot \exp\left\{j\frac{\pi}{K_a}(f_a - f_{d0})^2\right\} \exp\left\{-j\frac{\pi\hat{\alpha}}{K_a^3}(f_a - f_{d0})^3\right\} \\ & \cdot \exp\left\{j\frac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{4K_a^5}(f_a - f_{d0})^4\right\} \end{aligned} \tag{40}$$

where $W_a(\cdot)$ represents the frequency-domain window function. Moreover, the spectrum of the perturbed signal can be observed to shift to the position of the extra Doppler center, which is caused by the NCS. To facilitate the analysis of the spectrum, we reorganize the phase terms and ignore the constants and the first-order terms as follows:

$$\begin{aligned} \Phi_{a2}(f_a) = {} & \mu_2(f_{d0}) f_a^2 + \mu_3(f_{d0}) f_a^3 \\ & + \frac{\pi}{K_a} f_a^2 - \frac{\pi\hat{\alpha}}{K_a^3} f_a^3 + \frac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{4K_a^5} f_a^4 \end{aligned} \tag{41}$$

$$\begin{cases} f_{d0} = \dfrac{3}{2}\hat{\alpha}t_0^2 + 2\hat{\beta}t_0^3 \\ \mu_2(f_{d0}) = \dfrac{3\pi\hat{\alpha}}{K_a} f_{d0} + \dfrac{27\pi\hat{\alpha}^2 + 12\pi\hat{\beta}K_a}{2K_a^5} f_{d0}^2 \\ \mu_3(f_{d0}) = -\dfrac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{K_a^5} f_{d0}. \end{cases} \tag{42}$$

In (41) and (42), the last three terms in the phase are all invariant and can be compressed consistently. The first two terms, however, are influenced by the Doppler center, exhibiting Doppler-variant features. Therefore, we employ a third-order reference filter to perform further compensation

$$H_3(f_a) = \exp\left\{-j\pi\left[\mu_2\left(f_{d0}^{ref}\right) f_a^2 + \mu_3\left(f_{d0}^{ref}\right) f_a^3\right]\right\} \tag{43}$$

where f_{d0}^{ref} represents the referred Doppler center frequency, calculated from the referred azimuth time in each subimage. After filtering, the residual phase can be expressed as follows:

$$\Phi_{a3}(f_a) = \frac{\pi}{K_a} f_a^2 - \frac{\pi\hat{\alpha}}{K_a^3} f_a^3 + \frac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{4K_a^5} f_a^4 + \theta_{res}(f_a) \tag{44}$$

$$\begin{aligned} \theta_{res}(f_a) = {} & \left[\frac{3\pi\hat{\alpha}}{K_a} + \frac{27\pi\hat{\alpha}^2 + 12\pi\hat{\beta}K_a}{2K_a^5}\left(f_{d0} + f_{d0}^{ref}\right)\right]\Delta f_{d0} f_a^2 \\ & - \frac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{K_a^5}\Delta f_{d0} f_a^3. \end{aligned} \tag{45}$$

According to the above expressions, the main Doppler-variant terms are eliminated, and the residual phase error exhibits little variation, which will be further compensated by the following filter in the Doppler domain:

$$H_4(f_a) = \exp\left\{j\pi\left(\varepsilon f_a^2 + \gamma f_a^3\right)\right\}. \tag{46}$$

Finally, an invariant-parametric filter incorporating all known parameters can be applied to perform consistent azimuth compression directly

$$H_5(f_a) = \exp\left\{-j\left(\frac{\pi}{K_a} f_a^2 - \frac{\pi\hat{\alpha}}{K_a^3} f_a^3 + \frac{9\pi\hat{\alpha}^2 + 4\pi\hat{\beta}K_a}{4K_a^5} f_a^4\right)\right\}. \tag{47}$$

Because the target is noncooperative, most parameters in the proposed method cannot be calculated in advance, so multiple parameters must be estimated. In summary, the problem of azimuth-variant phase compensation can be formulated as an optimization involving four parameters

$$\left(\hat{\alpha}, \hat{\beta}, \hat{\varepsilon}, \hat{\gamma}\right) = \arg\min\left\{IE\left(I(m,n); \alpha, \beta, \varepsilon, \gamma\right)\right\}. \tag{48}$$

However, due to the high computational cost and limited solution accuracy, joint optimization of all four parameters simultaneously is not feasible. Fortunately, there exist heuristic algorithms for solving multiparameters' optimization problems, such as particle swarm optimization (PSO) [43] and genetic algorithm (GA) [44]. These nongradient methods circumvent the complex computation of the gradient in the transform domain and guarantee both accuracy and efficiency. Taking PSO as an example, the algorithm can reduce the search complexity to $O(P \cdot L \cdot F)$, where P, L, and F denote the particle swarm size, the iterative number, and the computational cost of the loss function (entropy), respectively. Specifically, the computational cost of the loss function in NCS is $O(2MN + MN\log M)$. Thus, the total cost of azimuth-variant phase compensation in all the subimages under the PSO framework is $O(2PBLMN + PBLMN\log M)$. In this article, we set the particle swarm size to 72. Combined with the calculated cost of MIME, the overall computational cost of SVPC could be expressed as $O(177BLMN + 73BLMN\log M)$.

After azimuth-variant phase compensation is performed on each subimage, the overlapping areas in the azimuth direction are first cropped to the predefined overlap size. Subsequently, the refocused subimages are mapped back to their corresponding azimuth and range bins determined during the partition process. Finally, all subimages are mosaicked to reconstruct the final refocused SAR image, thereby preserving the continuity and structural integrity of the ship target.

IV. Experimental Results

A. Numerical Simulation

To validate the effectiveness and analyze the performance of the proposed method, we conduct numerical simulations of a C-band single-channel spaceborne SAR operating in the side-looking mode. The key parameters of the radar signal are listed in Table I.

The simulated target is represented by a 7 × 3 scatterer array in Fig. 7(a). The target measures 150 m in length and 30 m in width, comparable to those of medium-sized vessels. To recreate the ship motion under moderate-to-rough sea conditions, we configured a coupled motion state involving both rotation and translation, which is a typical pattern in real maritime scenarios. The specific parameters are summarized in Table II. In addition, we set special scatterers P_1–P_4,

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TABLE I
PARAMETERS OF THE SIMULATED SAR SYSTEM

Parameter	Value
Centroid Frequency	5.400012 GHz
PRF	3716.467773 Hz
Antenna Aperture	5.0 m
Bandwidth	240 MHz
Pulse Width	55.0 us
Satellite Velocity	7569.846159 m/s
Scenery Center Slant Range	1030766 m

TABLE II
MOTION PARAMETERS OF THE ARRAY TARGET

Motion	Parameters	Values
Translational Motion	Steering angle (°)	30
	Velocity (m/s)	5
	Acceleration (m/s^2)	0.5
Yaw	Amplitude (°)	1.5
	Period (s)	9
Pitch	Amplitude (°)	0.9
	Period (s)	8
Roll	Amplitude (°)	5.0
	Period (s)	15

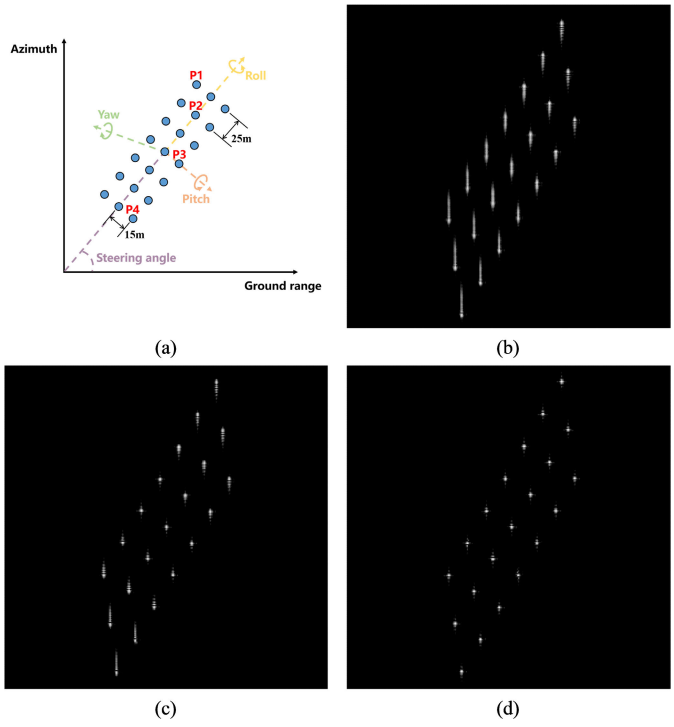


Fig. 7. Schematic and imaging results of the array target. (a) Schematic of ship structure. (b) Imaging result after RDA. (c) Imaging result after azimuth-invariant phase compensation. (d) Imaging result after SVPC.

representing the different sections on the ship, and specifically observe their refocusing results.

Fig. 7(b) shows the imaging results of a moving ship based on RDA. An obvious space-variant defocusing effect can be observed in the image, and the inherent translational motion can worsen the level of spatial variance in some cases. In demonstrating the SVPC processing, we present the intermediate result after azimuth-invariant phase compensation in Fig. 7(c) and the final refocused image of the proposed method in Fig. 7(d). It can be observed that the uniform component of defocusing energy has been effectively compensated in Fig. 7(c). Specifically, the rotation center, which is only influenced by translational motion, has been refocused into a point scatterer, confirming the effectiveness of azimuth-invariant phase compensation. In Fig. 7(d), the space-variant defocusing energy has been completely accumulated at each point scatterer at the Doppler center after the complete SVPC processing.

Next, we select five different algorithms for comparison with the proposed method, including PGA [8],

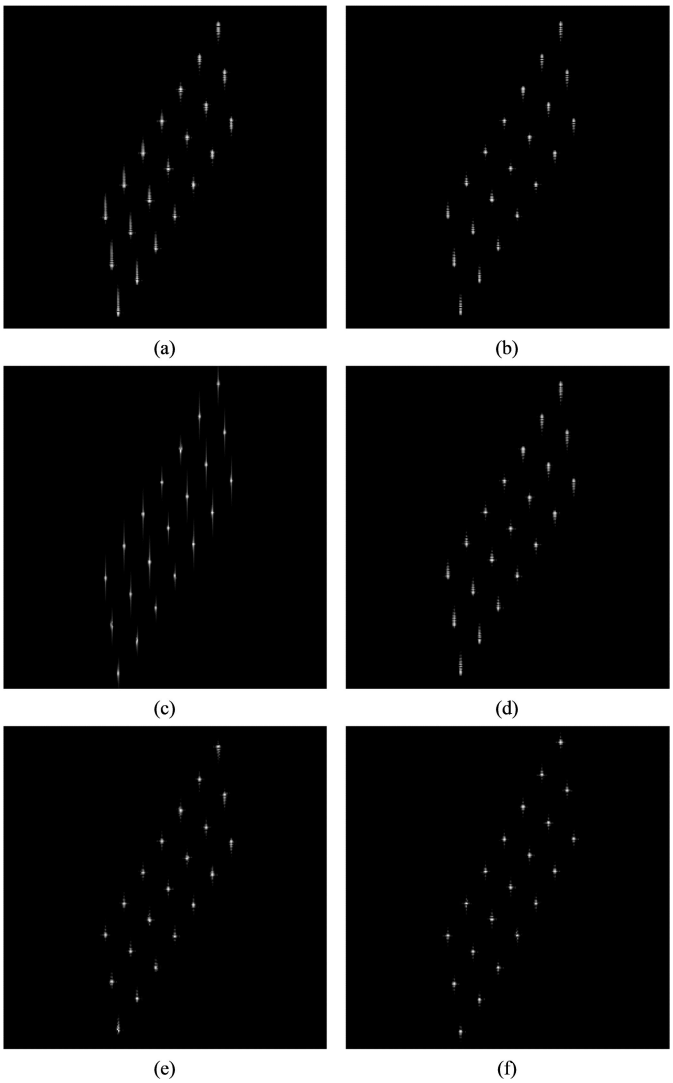


Fig. 8. Processed SAR images of the array target by comparison methods. (a) PGA. (b) LVD. (c) FrFT-WVD. (d) TMPR. (e) FNCS. (f) Proposed method.

Lv's distribution (LVD) [36], fractional Fourier transform-Wigner–Ville distribution (FrFT-WVD) [41], target motion phase refocusing (TMPR) [12], and frequency-domain NCS (FNCS) [23]. These methods fall into two categories: ISAR imaging and SAR imaging. In addition, the translational motion compensations of LVD and FrFT-WVD are both realized by average range profile entropy minimization (ARPEM) [25] and fast minimum entropy phase compensation

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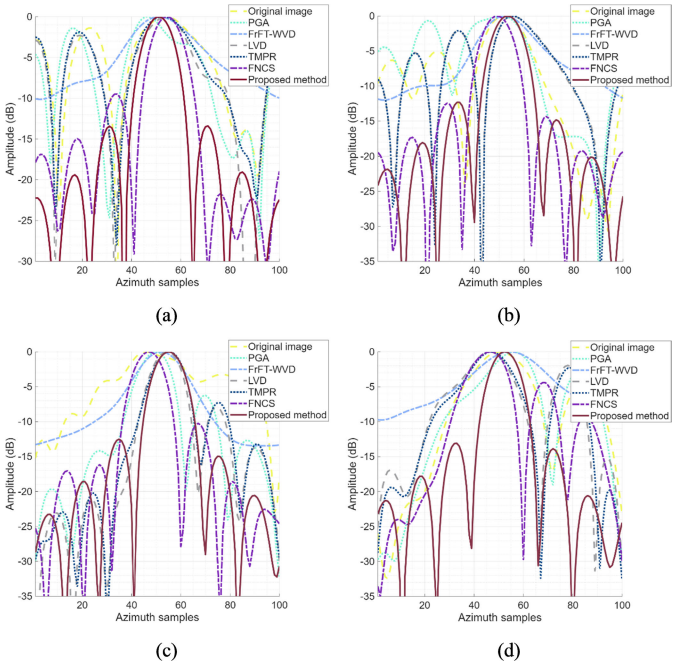


Fig. 9. Azimuth profiles of selected scatterers. (a) P_1. (b) P_2. (c) P_3. (d) P_4.

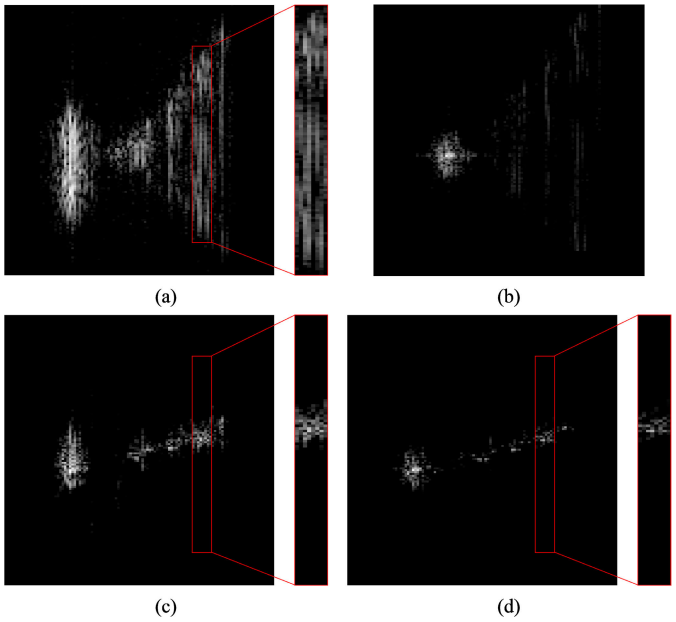


Fig. 10. Processed results of ship 1 by SVPC. (a) Original image. (b) Without image segmentation. (c) After azimuth-invariant phase compensation. (d) After complete SVPC processing.

TABLE III

MEASUREMENT RESULTS OF AZIMUTH PROFILES IN NUMERICAL SIMULATION

Scatterer	Method	Performance Index	
		PSLR (dB)	ISLR (dB)
P_1	Original image	-1.3603	-2.5538
	PGA	-1.3760	-3.0014
	LVD	-2.2860	-1.7630
	TMPR	-1.8651	-2.5797
	FNCS	-9.4792	-9.5480
	Proposed method	-13.3986	-11.7227
P_2	Original image	-5.0418	-4.3717
	PGA	-0.6660	-2.2751
	LVD	-2.1199	-3.0124
	TMPR	-2.1056	-2.9780
	FNCS	-12.4405	-11.1563
	Proposed method	-12.3056	-11.5690
P_3	Original image	-3.2412	-1.8884
	PGA	-6.2459	-6.5400
	LVD	-7.9187	-8.5591
	TMPR	-7.2623	-8.1427
	FNCS	-10.2557	-10.5025
	Proposed method	-12.5150	-11.9468
P_4	Original image	-4.4454	-6.6037
	PGA	-2.7575	-5.1527
	LVD	-1.9495	-4.6848
	TMPR	-2.2825	-5.0093
	FNCS	-4.3710	-5.6763
	Proposed method	-13.0808	-11.5943

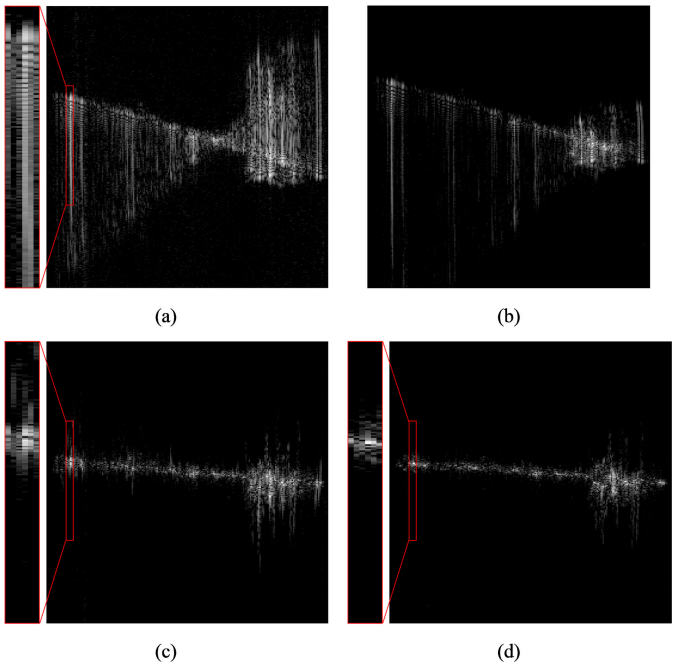


Fig. 11. Processed results of ship 2 by SVPC. (a) Original image. (b) Without image segmentation. (c) After azimuth-invariant phase compensation. (d) After complete SVPC processing.

(FMEPC) [37]. We also utilize the range segmentation and PGA technique in the FNCS method to eliminate the azimuth-invariant phase. The processed results of the simulated data by different methods are shown in Fig. 8(a)–(f).

The processed result of PGA is shown in Fig. 8(a) and is similar to the azimuth-invariant phase compensation algorithm proposed in this article. Only the invariant part of the defocusing energy can be corrected, while the space-variant blurring still degrades the image quality. The result of LVD in Fig. 8(b) shows a certain improvement compared with PGA. However, due to the limited accuracy of rotation velocity estimation under azimuthal variance, this method cannot fully compensate for complex spatial variance. The FrFT-WVD method utilizes FrFT to extract the LFM components and accumulates a high-resolution WVD without cross-terms, thereby overcoming the variant Doppler history caused by rotational motion. However, this quadratic distribution amplifies the energy difference between dominant and weak scatterers and leads to severe smearing. This blurs the ship structure and degrades the image

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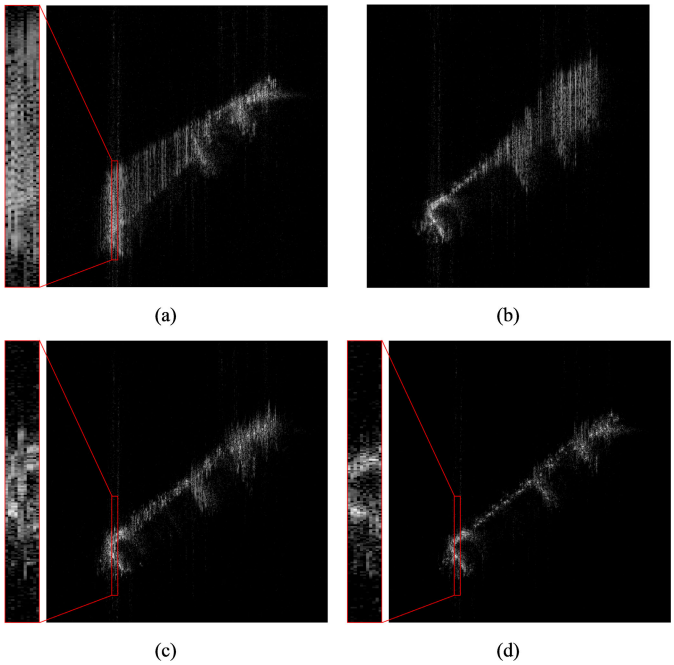


Fig. 12. Processed results of ship 3 by SVPC. (a) Original image. (b) Without image segmentation. (c) After azimuth-invariant phase compensation. (d) After complete SVPC processing.

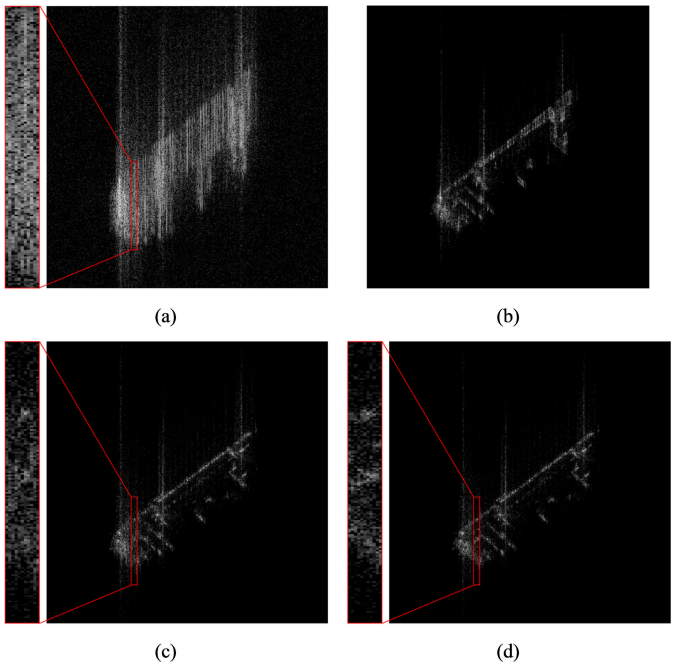


Fig. 13. Processed results of ship 4 by SVPC. (a) Original image. (b) Without image segmentation. (c) After azimuth-invariant phase compensation. (d) After complete SVPC processing.

quality, as shown in Fig. 8(c). The processed result of TMPR is shown in Fig. 8(d). This algorithm estimates the vessel's translational and rotational motions to compensate for the space-variant phase error, but it cannot handle the azimuthal variance of multiple components. The image processed by FNCS is shown in Fig. 8(e). Similar to the time-domain NCS, the method provides some suppression of azimuthal variance. However, under strong spatial variance, it fails to achieve complete refocusing. The SAR image processed by the proposed method is shown in Fig. 8(f). With the aid of image segmentation and time-domain NCS, the spatial variance is completely suppressed, and all scatterers are refocused at their Doppler centers. In summary, the proposed method achieves the best refocusing effect.

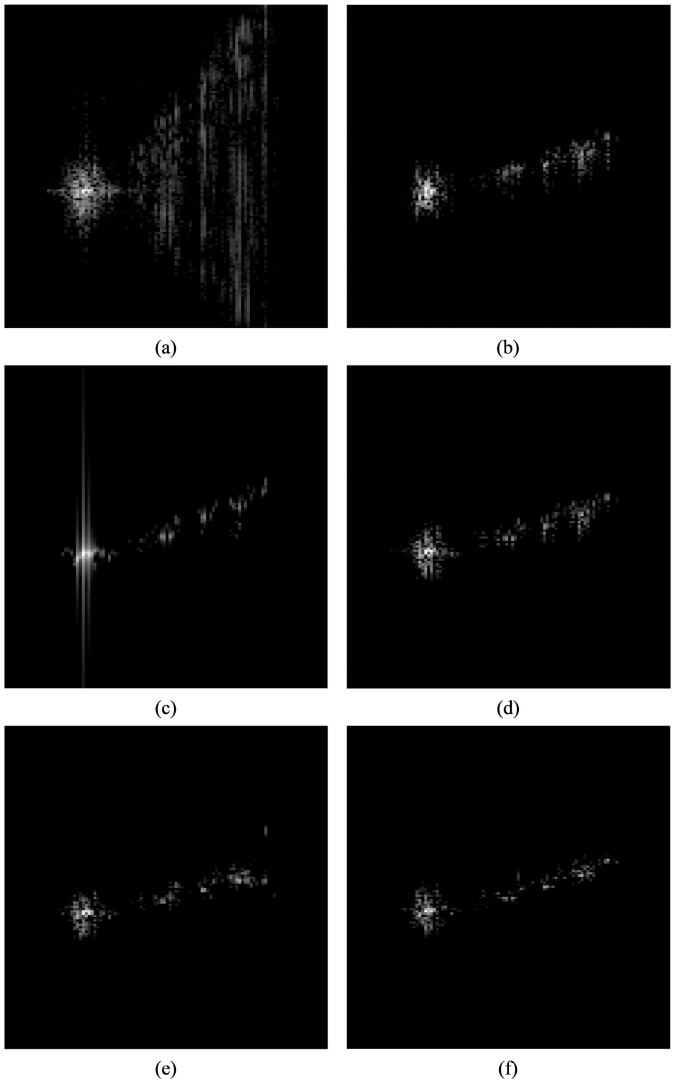


Fig. 14. Processed SAR images of ship 1 by comparison methods. (a) PGA. (b) LVD. (c) FrFT-WVD. (d) TMPR. (e) FNCS. (f) Proposed method.

Specifically, the azimuth profiles at selected scatterers (P_1–P_4) in processed results of different methods are shown in Fig. 9(a)–(d). The proposed method achieves the best performance compared with other algorithms. The azimuth sidelobes of each point scatterer are suppressed to a level that ensures effective imaging. Specifically, the peak sidelobe ratios (PSLRs) are all less than −12 dB, and the integration sidelobe ratios (ISLRs) are less than −11 dB, which represent the best values among the compared methods. The specific performance metrics of the algorithms are listed in Table III.

B. Real Spaceborne SAR Data

This section presents the performance of the proposed method on real spaceborne SAR data from the Umbra open dataset. The main parameters of the Umbra radar are listed in

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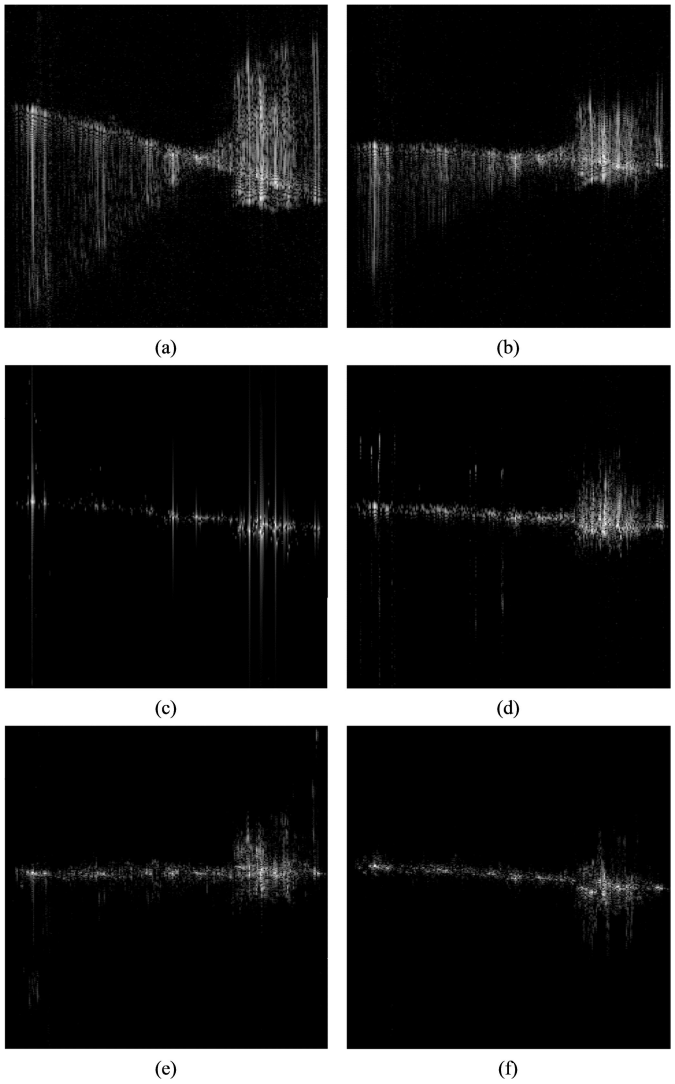


Fig. 15. Processed SAR images of ship 2 by comparison methods. (a) PGA. (b) LVD. (c) FrFT-WVD. (d) TMPR. (e) FNCS. (f) Proposed method.

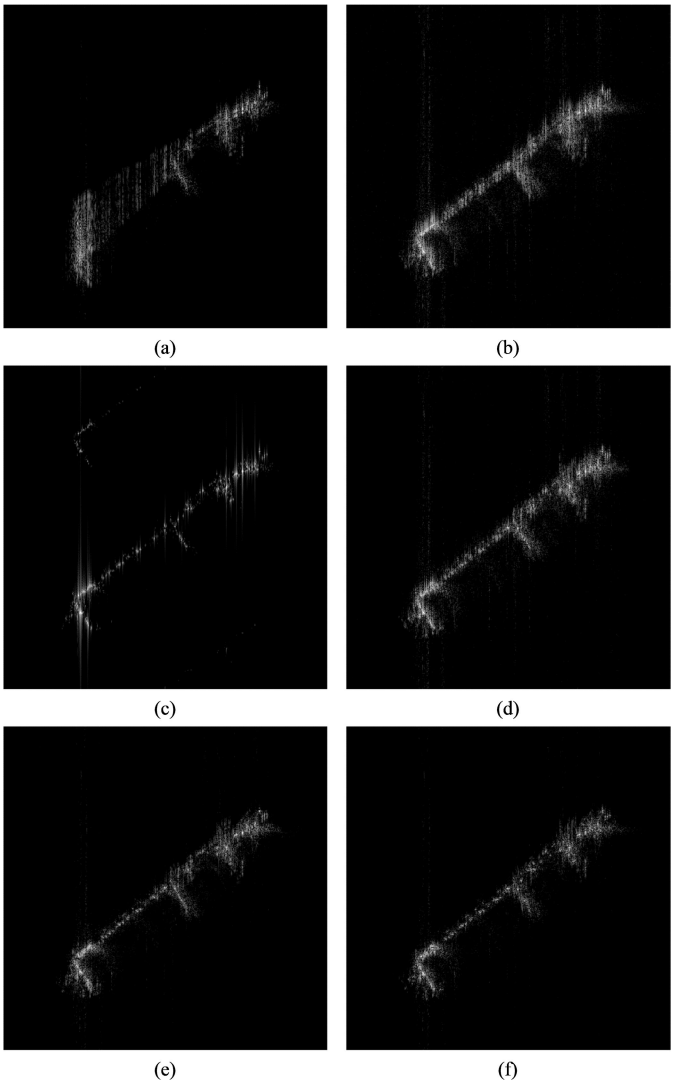


Fig. 16. Processed SAR images of ship 3 by comparison methods. (a) PGA. (b) LVD. (c) FrFT-WVD. (d) TMPR. (e) FNCS. (f) Proposed method.

TABLE IV
KEY PARAMETERS OF THE UMBRA RADAR SYSTEM

Satellite Velocity	7670 m/s
Center Frequency	9.80 GHz
PRF	3600 Hz
Bandwidth	1200 MHz
Orbit Height	517 km
Operation Mode	Spotlight

Table IV. The defocused ship images are extracted from the single-look complex (SLC) data of each system and processed separately. Specifically, for each target, we manually shift the image to the azimuth center of the entire scene to eliminate possible errors caused by the aperture center time and then crop it to a smaller size to accelerate processing.

Figs. 10–13 illustrate the results of the proposed SVPC method. The red boxes alongside the results show the processing effects on representative subimages after each step of the proposed method. Among them, Figs. 10(a)–13(a) show the defocused ships in the original SAR images, which exhibit severe space-variant defocusing and pose significant challenges for ship imaging. Figs. 10(b)–13(b) present the ablation experiments on the image segmentation strategy. Image segmentation improves adaptability to spatial variance and guarantees effective refocusing. Finally, Figs. 10(c)–13(c) and 10(d)–13(d) illustrate the results of azimuth-invariant and azimuth-variant phase compensation, respectively.

Similar to the simulated target experiment, we also employ the aforementioned five methods to validate the processing effects, as shown in Figs. 14–17. Specifically, the results of PGA, LVD, FrFT-WVD, TMPR, FNCS, and the proposed method are presented in Figs. 14(a)–17(a), 14(b)–17(b), 14(c)–17(c), 14(d)–17(d), 14(e)–17(e), and 14(f)–17(f), respectively. As demonstrated in the comparison experiments, the proposed method achieves the best refocusing effects on the real data.

To quantitatively evaluate the performance of the methods, we employ several image quality metrics, including image entropy, image contrast, and peak-to-average power ratio (PAPR). The definition of image entropy is given in (19), and

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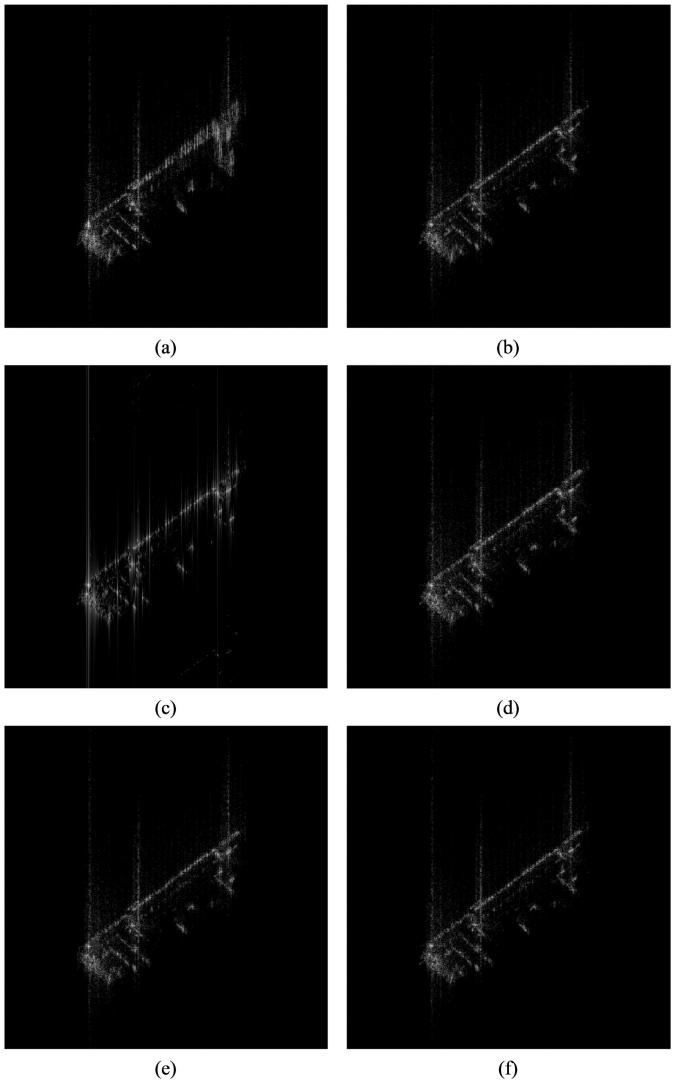


Fig. 17. Processed SAR images of ship 4 by comparison methods. (a) PGA. (b) LVD. (c) FrFT-WVD. (d) TMPR. (e) FNCS. (f) Proposed method.

the image contrast is defined as

$$IC(s) = \frac{\frac{1}{MN}\sum_{m,n}|s(m,n)|^2}{\left[\frac{1}{MN}\sum_{m,n}|s(m,n)|\right]^2}. \tag{49}$$

It measures the intensity difference between bright and dark regions in an image. Higher focusing quality leads to more distinct intensity variation and, thus, a higher contrast value. PAPR is a signal metric that reflects the prominence of peak power relative to the background. When defocused main lobes are refocused, the peak energy increases, resulting in a higher PAPR value. The expression for PAPR is given as follows:

$$PAPR(s) = 10\lg\left[\frac{\max\left\{|s(m,n)|^2\right\}}{\frac{1}{MN}\sum_{m,n}|s(m,n)|^2}\right]. \tag{50}$$

In summary, improved refocusing quality corresponds to lower image entropy, higher image contrast, and higher PAPR. The quantitative results for each dataset are summarized in Table V. Compared to the defocused SLC images, all the tested methods achieve measurable improvement in most cases. Among them, the proposed method delivers superior visual quality and achieves excellent scores across the evaluated metrics, which further validates its effectiveness and robustness. The proposed method incorporates both azimuth-invariant and azimuth-variant phase compensation for each subimage. Although MIME and heuristic optimization are adopted to alleviate the computational burden, iterative processing for large-scale real data remains computationally expensive. Hence, a tradeoff between refocusing accuracy and computational efficiency is inevitable. In practice, some preprocessing techniques, such as preestimation of the initial value and range, can be introduced to accelerate convergence and reduce the number of iterations.

TABLE V
PROCESSING PERFORMANCE OF DIFFERENT REFOCUSING METHODS IN REAL SPACEBORNE SAR DATA

Ship Index	Method	Image Quality		
		Image Entropy	Image Contrast	PAPR (dB)
1	Original image	7.3940	2.8764	20.3691
	PGA	6.3752	3.3606	31.1509
	LVD	6.2192	4.7198	27.5054
	FrFT-WVD	6.8227	3.3996	27.2792
	TMPR	5.8839	5.4397	29.2953
	FNCS	5.7687	5.4273	30.7218
	Proposed method	5.7393	5.4616	31.7331
2	Original image	9.7641	3.2627	31.4889
	PGA	9.4656	3.4044	35.4952
	LVD	9.3156	3.7045	32.6561
	FrFT-WVD	9.2242	6.4986	32.7815
	TMPR	8.6862	4.0416	34.8427
	FNCS	8.6038	4.1484	36.9076
	Proposed method	8.2467	4.4737	38.1945
3	Original image	10.004	3.2288	31.6310
	PGA	9.9497	3.2763	32.3635
	LVD	9.4999	3.7026	31.7616
	FrFT-WVD	9.9186	3.6274	30.0694
	TMPR	9.1609	4.3541	32.9870
	FNCS	9.1469	3.9213	34.8421
	Proposed method	8.9757	4.0389	35.0739
4	Original image	10.437	2.8055	26.2905
	PGA	9.7101	3.3844	35.4175
	LVD	9.2857	3.6189	39.8408
	FrFT-WVD	9.7012	4.4074	36.9414
	TMPR	9.2921	3.6494	39.0445
	FNCS	9.3081	3.5839	39.8997
	Proposed method	9.1732	3.6750	40.3008

V. CONCLUSION

This article presents an SVPC method for refocusing moving ships in SAR imagery. First, through approximate analysis and spatial variance modeling of nonstationary moving ships, we derive the space-variant phase signal in polynomial form and separate it into azimuth-invariant and azimuth-variant components. Image segmentation is then employed to reduce errors in spatial variance processing, and a fast iterative algorithm based on MIME is presented to compensate for the azimuth-invariant terms, in which an optimized transformation is introduced to replace the computationally intensive enumeration and ensure the accuracy of parameter estimation. After that, we propose an azimuth-variant phase compensation method via subimage NCS. The main processing steps include time-domain perturbation, modified Doppler correction, and consistent compression. This approach suppresses azimuthal

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variance to the greatest extent. Experiments on both simulated and real SAR data validate the effectiveness and robustness of the proposed method.

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